

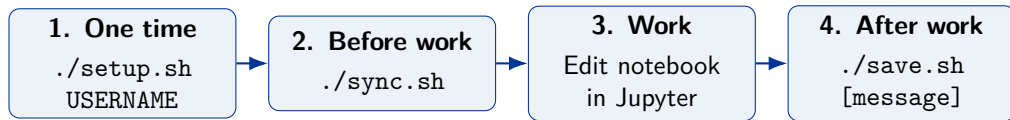
ENGR 1451/2451: Exploratory Data Science

Simplified Git Workflow with Setup, Sync, and Save Scripts

Prof. Paul W. Leu
University of Pittsburgh

Fall 2026

The Whole Workflow: Three Commands



The scripts are meant to hide most Git complexity

Students still need the big idea: **main** is for instructor material; **student-work** is for coursework.

Important Rule

Do not do coursework on main

main should stay a clean copy of the instructor repository.

main

- receives instructor updates
- should stay clean
- should not contain completed notebooks

student-work

- your branch for coursework
- where notebooks are edited
- where commits should happen

Step 1: One Time Only

First fork the instructor repository on GitHub:

```
https://github.com/pleu/26f-engr1451-2451
```

Then run the setup script:

```
./setup.sh YOUR_GITHUB_USERNAME
```

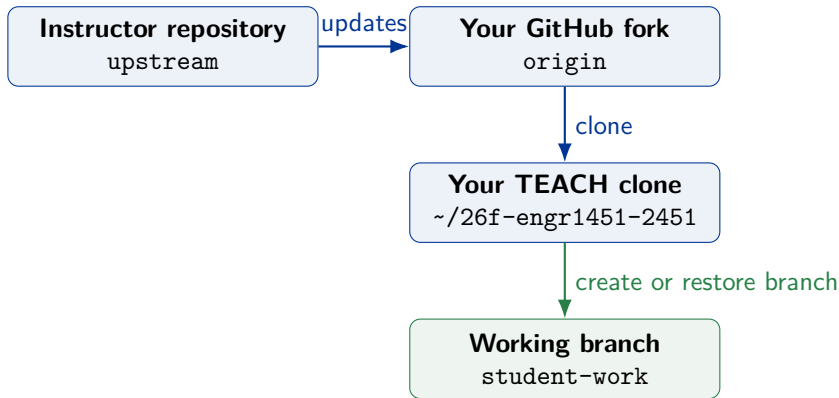
Example

```
./setup.sh jdoe
```

Repository folder

```
~/26f-engr1451-2451
```

What setup.sh Sets Up



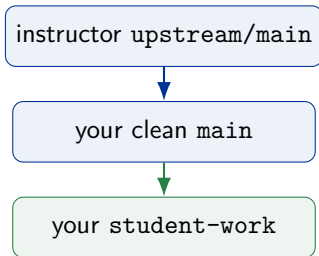
After setup

main tracks instructor updates; student-work is created or restored for your commits.

Step 2: Before Starting New Work

Run the sync script before starting a new assignment or class activity:

```
./sync.sh
```



What `sync.sh` does

Gets instructor files, updates main, then merges them into student-work.

What `sync.sh` Protects

Before syncing

- checks for unsaved changes
- stops if work needs to be saved first
- avoids mixing unfinished edits into a sync

While syncing

- updates `main` from instructor repo
- refuses messy updates to `main`
- switches back to `student-work`

If `sync.sh` stops

Do not force anything. Run `./save.sh` first or ask for help.

Step 3: After Working in Jupyter

Save the notebook in Jupyter, then run:

```
./save.sh
```

Optionally add your own commit message:

```
./save.sh "Complete w03a exercises"
```

What `save.sh` does

- checks that you are on `student-work`
- shows changed files and asks for confirmation
- commits with your message or a timestamp
- pushes to your GitHub fork

Commit message is optional: use your own message or the timestamped default.

The Safety Check in `save.sh`

Allowed



Commit on `student-work`

Blocked



Commit on `main`

What students should remember

If `save.sh` says you are on `main`, switch back to `student-work` before saving.

Student Command Summary

One time only

```
./setup.sh YOUR_GITHUB_USERNAME
```

Before starting new work

```
./sync.sh
```

After working in Jupyter

```
./save.sh  
# or: ./save.sh "Complete w03a exercises"
```

What this means

- `setup.sh`: create repo; restore branch if needed
- `sync.sh`: get new instructor files
- `save.sh`: commit and push work

Rule

Do not do coursework on main.

Common Messages

Repository already
exists

Setup was already done. Use `./sync.sh` instead.

Existing student-work
found

Setup found your old work branch on GitHub and re-stored it locally.

Run `setup.sh` first

The repo was not found in `~/26f-engr1451-2451`.
Complete setup.

You have unsaved changes

Run `./save.sh` before syncing.

You are on 'main'

Switch to student-work before saving coursework.

merge conflict

You and the instructor changed the same file. Stop and ask for help.

What to Submit

GitHub backs up your work

The scripts push your work to the `student-work` branch of your GitHub fork.

Canvas is the official submission

Submit the saved `.ipynb` notebook in Canvas, following the assignment instructions.

Final check

Open your fork on GitHub, select the `student-work` branch, and confirm your notebook appears there.